

# Categorizing Open Government Data Users by Exploring their Challenges and Proficiency

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## ABSTRACT

Developing user personas has been a crucial aspect of human-centered system design for decades as it helps in understanding and segregating users based on their prominent characteristics. However, such a technique has not been applied in developing and improving systems for supporting open government data (OGD) users. Therefore, this paper explores OGD users' characteristics and creates relevant personas for them. Open coding-based content analysis and k-means clustering were performed on posts of an online community managed by a U.S. local-level OGD portal, where users' characteristics such as purposes, challenges, and proficiency in civic data domain knowledge and computational skills were used as features for clustering. Through manually analyzing the output clusters, we identified three personas with their distinct behavior patterns based on their proficiency. The proposed personas can facilitate in personalizing informational and technological support for OGD forum users.

## CCS CONCEPTS

• **Human-centered computing** → **HCI theory, concepts and models.**

## KEYWORDS

persona, Open Government Data users, human-centered systems, online communities

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## 1 INTRODUCTION

Since around 2009, a massive amount of government data, from federal to local levels, has been made available via various open government data (OGD) portals. The value and impact of OGD have been extensively discussed and confirmed by existing studies. Through OGD recognition, the transparency and accountability of

public bodies are improved. Studies show that efficient and effective usages of OGD can promote social engagement, citizen-centric services, business creation and innovation, and great economic value [1, 14, 28]. Nonetheless, the premise that these values can be achieved relies on the assumption that the data can be effectively and widely used. However, studies discovered that users still often experienced difficulties in interacting with OGD, such as finding data [19]. In a survey of 3,212 participants, Pew Research Center found that even though people of the USA are optimistic about OGD, very few think that “agencies are doing a great job of providing useful data,” and few respondents considered the government as being “very effective” at sharing data with the public [2].

To address these problems, we explored the characteristics of OGD users in order to further support users based on developed personas. Developing stereotypes of users called *personas* has been consistently studied by HCI researchers and practitioners in various fields, such as system designers and healthcare [25]. Personas segregate users into groups based on their prominent characteristics thus promoting a deep understanding of user populations. Investigating user characteristics has been recognized as a crucial and effective aspect of human-centered system design for decades [11]. In this paper, we want to fill the existing gap in the OGD domain and introduce OGD user personas by categorizing them based on their challenges and proficiency in using data. Through developed personas, we can study the specific needs of each user, facilitate OGD portals in the improvement of their system design, and deliver customized informational support to the users.

Adopting online community data to study users has been broadly used in the health field [15, 32]. This is because online forums contain abundant and unobtrusive interactions among users, where they can post requests, questions, and answers, comment on others' posts, or simply socialize with other users. The forums also provide rich user-generated content that presents users' motivations, challenges, and stories, all of which reveal their interacting behaviors. Consequently, we used the online posts from an OGD discussion group to evaluate OGD users' characteristics. This forum (<https://groups.google.com/g/opendataphilly>) is managed by *OpenDataPhilly* (a US local-level OGD portal), and it allows the public to ask questions about the location of specific data, data interpretation, as well as other challenges encountered.

Our study extracts OGD users' characteristics and then groups similar users via clustering so that each cluster acts as a persona. Accordingly, the main research question of this study is *what are the OGD users' characteristics and the corresponding OGD user personas on an OGD forum?*

The identified personas and the corresponding characteristics will inform OGD portal's administrators and system designers to

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better understand their users and provide customized support to the users. Our study is novel because it is the first attempt to identify personas for OGD users with both qualitative and quantitative methodology, and it could be expanded further with more complex personas and more extensive user behavior investigation. These methods can inspire HCI researchers to further study OGD users.

## 2 LITERATURE REVIEW

**Research studies concerning OGD users.** With the development of OGD, an increasing number of researchers and OGD practitioners agreed that understanding users will contribute to improving the usage of OGD data. The extant studies on OGD users focused on the investigation of stakeholders [7, 14], user behaviors [6, 17, 19, 31], user challenges [13, 33], and user needs [8]. Regarding user behaviors, Koesten et al. [19] employed an in-depth semi-structured interview method to understand user behaviors when interacting with OGD, and then formulated a framework for human structured-data interaction. Additionally, there are studies using a transaction log analysis to study OGD user behaviors patterns, characteristics of issued queries, and data needs [17, 31]. All the mentioned studies have provided information about users, but few studies explored the characteristics of OGD users that are critical to understanding users within the HCI field.

Also, few research studies leveraged the online discussion community data to understand users in the OGD domain. We believe that online discussion community data is rich, user-generated, and unobtrusive data, which can be useful for extracting user characteristics.

**Persona development.** Persona development is accepted as a critical process to understand users in a particular domain [5, 23]. Availability of large user data through users' online interactions and activities became the driving force for the data-driven persona generation [22]. These persona development approaches were either used independently or combined with manual analysis to provide more insights from domain experts. Approaches in unsupervised data-driven persona development can be classified into two major groups: dimensionality reduction-based methods and clustering-based methods. Dimensionality reduction-based methods map users in a lower dimension space [3, 12, 26]. The drawback of a dimensionality reduction-based method is that it is complex for the layman to understand, which makes it hard to interpret the persona output. These concerns caused clustering-based methods to become popular for persona development. Clustering-based methods have been applied in various domains, including health [16], recommendation system [26] and education [9], to name a few. A clustering-based method groups the user-data based on similarity metrics, which is qualified to our study.

## 3 METHODOLOGY

To answer the research question, both qualitative and quantitative methods were adopted in this study. We first applied content analysis to explore the user characteristic features. Also, a k-means clustering algorithm was performed to classify OGD users. Through a chi-squared test, we extracted significant features in each cluster, and then manually analyzed and understood the characteristics of each cluster, and finally grouped clusters to create personas.

### 3.1 Data Collection

Our research was conducted on the public online forum of *OpenDataPhilly*, where users can freely join the group and start new threads or comment on others' existing threads. The data was collected from January 2019 to June 2021, containing 174 threads, spanning 505 posts by 155 users. Also, since this study attempts to examine the OGD users in this forum, we decided to exclude the data from the 3 portal administrators, including their initiated posts and replies. Consequently, we obtained 152 users and 338 posts. This work was approved by our university's Human Research Protection Office.

### 3.2 Identifying Characteristic Features

In our effort to group OGD users, we needed to identify OGD users' characteristics. Since this study is the first one to explore OGD user personas, we decided to focus on the most essential characteristics related to OGD use. We first identified the *purposes (post types)* presented in the post. This feature demonstrates the effort users put into their posts. Next, we proposed the second feature: *challenges (question categories)*. By exploring user challenges when interacting with OGD and the portal, the barriers caused by both the portals and users were revealed. Finally, we identified two remaining features: the degree to which they are proficient in *civic data domain knowledge (CDDK)* and *computational skills (CS)*. One of the characteristics of OGD is that to use OGD requires a certain level of data literacy [27]. Integrating these features enables us to sketch various personas for OGD users. The elaboration of the four features and related coding schema is presented as follows.

**Feature: Purpose.** In this work, *purpose* refers to the motives of users while posting on a forum, for instance, asking or answering a question. We referenced Kim et al.'s dialogue act (DA) tagging to create the coding schema to label the posts [18]. This study claimed that this post label set aims to capture the interactions that happen in troubleshooting-oriented threads, which is a satisfactory reference to construct our coding schema based on our data. Table 1 is the illustration of the coding schema.

**Feature: Challenge.** In this study, *challenge* denotes the topics of a question asked. For example, a user encounters difficulties in finding the sources of a dataset or understanding the meaning of column headings. To categorize the challenges, we adopted Xiao et al.'s [30] Framework for Structured Research Data Interaction as the basis for creating the coding structure of user challenges because we attempted to identify user challenges based on their interacting behaviors. We further modified Xiao et al.'s framework according to our demands to finalize the coding schema. Detailed explanations of the categories are provided in Table 2.

**Feature: The Proficiency of Computational Skills (CS).** In this study, *the proficiency of CS* indicates the displayed ability to use computational technologies, including computational tools (Excel, Tableau) or programming language (SQL, Python), to process data and to solve problems. Regarding the user proficiency levels, we referenced Lui and Baldwin's [21] idea that classified the proficiency into five levels. Through comprehensive discussions between authors, we considered that three levels are enough to represent OGD users' displayed proficiency, so we determined to classify the displayed

**Table 1: Code schema for user purposes**

| Code                   | Description                                                                                                                                         |
|------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------|
| Question               | The post consists of a question(s) from a user. This category includes initial questions and adding questions from the same user or other users.    |
| Question-Clarification | The post supplements more details to clarify the previous question.                                                                                 |
| Answer                 | The post provides an answer to a question or a report. This category contains initial answers and adding answers from the same user or other users. |
| Follow up Action       | The post claims to take follow up actions to make some improvements for the data to help users. For example, to learn new tools.                    |
| Report                 | The post reports an issue about the data, simply letting others know the current situation, not expecting to obtain answers.                        |
| Request                | The post request to publish a new dataset that is not available in the portal.                                                                      |
| Courtesy               | The post shows appreciation to answers.                                                                                                             |
| Other                  | The post does not belong to any of the above classes.                                                                                               |

**Table 2: Coding schema for user challenges**

| Code       | Description                                                                                                                                                                                     |
|------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Find       | The post presents the question when users encountered the challenges of finding the data. For example, asking help about where or how to find data sources.                                     |
| Assess     | The post presents the question when users critically assessed if the identified dataset is usable for their project, e.g. examining the file formats and specific data types.                   |
| Acquire    | The post presents the question when users acquired the datasets before or after assessing the data, such as clicking the provided link to download the datasets, or querying data through APIs. |
| Understand | The post presents the question when users encountered challenges of understanding data, e.g., asking the data collecting process.                                                               |
| Schematize | The post presents the question when the user cannot understand the logic of the data to establish the relationships between data.                                                               |
| Use        | The post presents the question when users cannot use the data for different reasons.                                                                                                            |
| Share      | The purpose of this post is to inform others under this thread or forum that the poster will share the data that they collected from various data sources by themselves.                        |

proficiency into three levels: novice, intermediate and expert. The details are demonstrated in Table 3.

*Feature: The Proficiency of Civic Data Domain Knowledge (CDDK).* Given the determination that proficiency in CDDK is a crucial aspect of using OGD [10], similar to the proficiency of CS we encoded the proficiency of CDDK into three levels (see Table 3).

During the encoding of the proficiency in CS and CDDK, when a user posted more than one post, we recorded the post with the highest level of proficiency as the user's proficiency. We acknowledge the coding method limitation since some posts may not have been able to demonstrate users' real proficiency in CS or CDDK. Therefore, we define the proficiency in this paper as "displayed proficiency." On the other hand, it is worthwhile to notice that for the users with high CS or CDDK, their questions were more highly likely to be more complicated, so it required them to provide a detailed description about what they already had and what support

they needed, which helped reveal their abilities. Also, this method has been used and evaluated by [21].

Based on the coding schema developed for the four features, a three main step content analysis was conducted. After developing an initial coding schema based on the selected features, the three authors discussed the schema together to refine the categories and descriptions, then the first two authors annotated the posts by using the finalized version code-book. The first author coded all the posts, and the second author randomly coded 30% posts. Cohen's Kappa [4] agreement scores for Purpose, Challenge, Proficiency in CS, and Proficiency in CDDK are 0.80, 0.88, 0.73, and 0.81, respectively. These agreement scores indicate a substantial agreement [29] between the two coders. We will further discuss the disagreements when we have more data in the future studies.

**Table 3: Coding schema for proficiency in CDDK and CS**

| Code                  | CDDK Description                                                                                                                                                                                                                   | CS Description                                                                                                                                                                                                                                                              |
|-----------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Novice (score1)       | The post of this user makes it patently obvious that s/he is not aware of the existence of a dataset or has no clue where to find the needed dataset.                                                                              | The posts of the users make it patently obvious that they have limited computational knowledge and/or understanding of the basic knowledge about data, such as file/data types.                                                                                             |
| Intermediate (score2) | The post of this user shows the ability to find the dataset's source, but s/he is not familiar with the context information or some key concepts about the needed datasets.                                                        | The posts of the users shows the familiarity of general computational knowledge, including downloading data (manually or using API) and using Excel and basic statistical knowledge to manipulate data.                                                                     |
| Expert (score3)       | The post of this user demonstrates the familiarity of the related contextual information about the dataset. The user gives the impression of knowing what they are talking about, with good insights into the topic of the thread. | The posts of the users present that they are good at data processing skills. For example, use programming languages to download and manipulate data. The user gives the impression of knowing what they are talking about, with good insights into the topic of the thread. |

### 3.3 Persona Implementation

K-means was selected for clustering as it has been shown to provide good insights to persona development [16, 20, 22]. We aggregated the features to the user level as our purpose is to explore the categories of users in the forum. Below are the details and motivation behind each feature aggregation.

*Purposes:* for each purpose, we counted the number of times the user posted with the same purpose. The hypothesis is that people with different posting purposes can provide us with insights into users' characteristics, e.g., users who provide more answers or users who ask lots of clarifications.

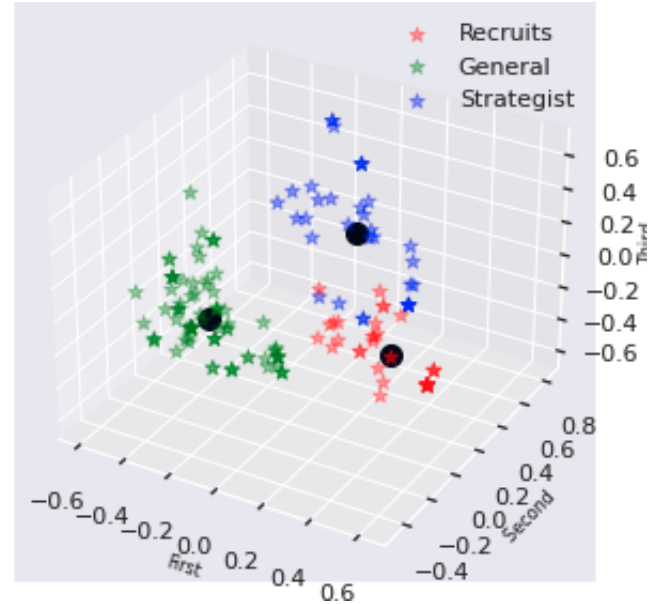
*Challenge:* we created a binary feature set for the *challenges* feature, which consisted of 1 if a user had a post with a particular challenge; otherwise the value of the feature was 0. We kept the *challenge* feature as binary, as we intended to know what kind of challenges different users had.

*Proficiency:* as proficiency is annotated based on users' displayed proficiency in a single post, we used the maximum score of the annotated proficiency across all the user's posts as the final score for the user.

We standardized the data by normalizing the value across each feature to scale from 0 and 1. K-means is an unsupervised algorithm that requires the number of clusters (K) as input. The Silhouettes score [24] was used to decide the number of clusters, which was three.

## 4 RESULTS

Three clusters were discovered in this study. Chi-squared test and post-hoc analysis were performed to find significant features in each cluster as shown in Table 4. All the reported p-values were Bonferroni corrected. We performed post-hoc analysis per features with groups as clusters, so the chi-squared values listed are associations between clusters and challenges, between clusters and purposes of posts, and between clusters and proficiency. Table 4 presents the cluster representation based on the features and the sum of feature values. The bold numbers are significant values in



**Figure 1: PCA based cluster visualization. Each dot represent a user in the OGD forum. The three black dots are cluster centroids**

the clusters and are provided along with their *Chi-squared* statistics and p-value. We plotted PCA based cluster components to visualize the three clusters, as shown in Figure 1.

**Cluster 1: General.** The users in this cluster have significantly low challenges related to *Acquire* ( $X^2 = -3.09, p = 0.0009$ ) and *Find* ( $X^2 = -2.76, p = 0.0028$ ), and they answer significantly more questions ( $X^2 = 4.41, p = 0.0000$ ) and ask significantly fewer questions ( $X^2 = -2.98, p = 0.0013$ ) in comparison to other two clusters. This cluster includes significantly more users with average or high CS ( $X^2 = 8.57, p = 0.0000$ ). Through manual analysis, we found that

**Table 4: Cluster validation with chi squared and p-values**

| Features               | Cluster 1                                   | Cluster 2                                  | Cluster 3                                 |
|------------------------|---------------------------------------------|--------------------------------------------|-------------------------------------------|
| <b>Challenges</b>      |                                             |                                            |                                           |
| Find                   | <b>13</b><br>( $X^2 = -2.76, p = 0.0028$ )  | <b>28</b><br>( $X^2 = 2.45, p = 0.0072$ )  | 20<br>1                                   |
| Assess                 | 11                                          | 5                                          | 9                                         |
| Acquire                | <b>22</b><br>( $X^2 = -3.09, p = 0.0009$ )  | 13                                         | 23                                        |
| Sensemaking            | 10                                          | 7                                          | 7                                         |
| Use                    | 10                                          | 3                                          | 2                                         |
| Share                  | 4                                           | 1                                          | 0                                         |
| <b>Purpose of post</b> |                                             |                                            |                                           |
| Questions              | <b>61</b><br>( $X^2 = 2.98, p = 0.0013$ )   | <b>73</b><br>( $X^2 = 3.28, p = 0.0005$ )  | 55                                        |
| QuestionsClarification | 5                                           | 1                                          | 5                                         |
| Answers                | 37                                          | <b>9</b><br>( $X^2 = -2.97, p = 0.0010$ )  | 11                                        |
| AnswerConfirmation     | 6                                           | 5                                          | 2                                         |
| Report                 | 4                                           | 0                                          | 1                                         |
| Request                | 3                                           | 2                                          | 6                                         |
| FollowUpAction         | 1                                           | 5                                          | 0                                         |
| <b>Proficiency</b>     |                                             |                                            |                                           |
| CDDK high              | 43                                          | <b>15</b><br>( $X^2 = -3.45, p = 0.0002$ ) | 56                                        |
| CDDK low               | 12                                          | <b>19</b><br>( $X^2 = 4.84, p = 0.0000$ )  | <b>7</b><br>( $X^2 = -3.89, p = 0.0000$ ) |
| CS high                | <b>45</b><br>( $X^2 = 8.57, p = 0.0000$ )   | <b>9</b><br>( $X^2 = -7.81, p = 0.0000$ )  | 0                                         |
| CS low                 | <b>10</b><br>( $X^2 = -7.641, p = 0.0000$ ) | 25                                         | <b>63</b><br>( $X^2 = 6.34, p = 0.0000$ ) |

78% (43/55) users in this group had average or high CDDK. Therefore, we intend to consider this group as the *General*. The main characteristics of this group are the users have more experience of using OGD. They ask high-level questions and are willing to answer others' questions using their high proficiency in CS and CDDK. Figure 2 shows the persona for *General* with example questions and answers from this group along with their proficiency.

**Cluster 2: Recruit.** Compared to other groups, the significant challenge in this group is *Find* ( $X^2 = 2.45, p = 0.0072$ ). People in this group ask significantly more questions ( $X^2 = 3.289, p = 0.0005$ ) and provide significantly less answers ( $X^2 = -2.97, p = 0.0010$ ). The significant test shows that the group has significantly more users with low CDDK ( $X^2 = 4.844, p = 0.0000$ ). To better understand the users in this cluster, we manually scrutinized all the records, and we found

that compared to the other clusters, there were more low CS users in this group ( $X^2 = -7.81, p = 0.0000$ ). Also, an interesting point is that all the answers in this group were provided by 4 (11%) users who are classified as high-level CS and CDDK. After examining all the records, we noticed that these 4 (11%) users were categorized in this group because although they were ranked as high-level CS and CDDK proficiency users, they asked more questions, which is a characteristic of this cluster. Given all these considerations, we recognize this cluster as *Recruit*. The main characteristics of this group are the users are newbies to OGD and OGD platforms. They have comparatively less proficiency in CS and CDDK, so they ask more questions and do not answer many questions. Moreover, their questions are more basic-level questions related to finding

|                                                                                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                                                |
|-----------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------|
|  | <b>General</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                                                |
| Challenges                                                                        | Data sensemaking, Data use and Data sharing                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                                                |
| Purpose of posting                                                                | Asking and Answering questions                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                                                |
| Answers                                                                           | <b>Be willing to answer others' questions.</b><br>Example answer: We worked with this data over the summer and composed a blog post which walks through a process to clean the data. You are correct the data are lacking IDs and have many duplicates. The blog post links to a GitHub repo which has some scripts that you may be able to modify for your purpose. I would also highly recommend contacting XXXXXXXX at the Board of Ethics, he is an expert on the data and the processes for submission which result in many of the issues you raised.                                                                                 |                                                |
| Questions                                                                         | <b>Asking higher level questions.</b><br>Example question: I've tried downloading a number of shp files for data from the City of XXXXXX that will not import to ArcGIS Pro through feature class to feature class. All of these files that fail are similar in that the file names are a string of numbers / letters / dashes. An error comes up saying that the input feature section "does not exist or is not supported." When I've renamed the file, I no longer get that error but when I try to run it, I get an error saying that it "cannot create output." I've triple checked the pathways. Any help would be much appreciated! |                                                |
| Proficiency                                                                       | Intermediate and High level civic data domain knowledge                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Intermediate and High level computation skills |

Figure 2: Cluster\_1 user persona: *General*

data. Figure 3 demonstrates persona of *Recruit* with some example questions and answers from this group along with their proficiency.

|                                                                                   |                                                                                                                                  |                              |
|-----------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------|------------------------------|
|  | <b>Recruit</b>                                                                                                                   |                              |
| Challenges                                                                        | Data find and Data acquire                                                                                                       |                              |
| Purpose of posting                                                                | Asking more questions                                                                                                            |                              |
| Answers                                                                           | Almost not answering questions                                                                                                   |                              |
| Questions                                                                         | <b>Asking basic level questions.</b><br>Example question: I am looking for a list of all of XXX's public parks. Does this exist? |                              |
| Proficiency                                                                       | Low level civic data domain knowledge                                                                                            | Low level computation skills |

Figure 3: Cluster\_2 user persona: *Recruit*

|                                                                                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                              |
|-------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|
|  | <b>Strategist</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                              |
| Challenges                                                                          | Data find and Data acquire                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                              |
| Purpose of posting                                                                  | Asking questions and answering less questions                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                              |
| Answers                                                                             | <b>Only answer civic data domain knowledge related questions.</b><br>Example answer: I remember the issue of officer identity came up a couple years ago in the context of research into stop and frisk complaints. Journalists had to do independent work to match those IDs with specific officers. Here is some coverage from the time, I believe things are still essentially the same: <a href="https://www.XXXXXX">https://www.XXXXXX</a> . POWER has done a lot of good research and analysis on this. <a href="https://XXXXX">https://XXXXX</a> . |                              |
| Questions                                                                           | <b>Asking basic level questions.</b><br>Example question: I am having trouble downloading both part one and 2 CSV files. Any guidance?                                                                                                                                                                                                                                                                                                                                                                                                                    |                              |
| Proficiency                                                                         | Intermediate and High level civic data domain knowledge                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Low level computation skills |

Figure 4: Cluster\_3 user persona: *Strategist*

**Cluster 3: Strategist.** There are no significant challenge or purpose for posting in this group. The users in this group have significantly more users with low CS proficiency ( $X^2 = 6.34, p = 0.0000$ ) than users in other groups. However, this group is not associated

with users with low CDDK proficiency ( $X^2 = -3.89, p = 0.0000$ ). Further analysis of this clusters revealed that not only was the prominent characteristics of this cluster low CS, but most of users had intermediate and high CDDK ( $56/63 = 89\%$ ). Manual analysis of the post content in this group disclosed that the answers provided by the users regarded the sources (locations) of data, which require CDDK proficiency. Hence, we call this group *Strategist*. *Strategist* refers to a user who know affluent civic data background knowledge, including knowledge about various data sources, and even know the workflows of some datasets. These skills provide them with the capability to answer CDDK related questions. However, they may not often use the data to perform tasks themselves, thereby they lack the CS proficiency. Figure 4 demonstrates the persona of *Strategist* with some example questions and answers from this group along with their proficiency.

## 5 DISCUSSION AND IMPLICATION

**Building an OGD online discussion community to support users.** In this work we utilized a k-means based clustering technique for persona generation as a first step to understand personas in OGD communities. Currently, very few OGD portals have an online forum, but our results demonstrate that establishing an online discussion community is beneficial to both users as well as OGD administrators. Users in the *General* group are willing to be helpful in answering questions and sharing data. Their answers enable other users to obtain access to data, so the overall effort and time for accessing and using OGD become less for new users. Users in the *Recruit* group are novices to OGD, but they have a desire to learn more. Through an online forum, *Recruit* users can fulfill their desire of learning via engaging with other users on the platform. *General* group users can be considered as supporters and *Recruiters* are information seekers. As for the *Strategist* group, even though they are not asking significant amount of questions nor answering many questions, they provided professional responses to CDDK-related questions. Given their CS levels, we assume *Strategists* would enjoy learning new CS skills to use data while answering the questions with their CDDK expertise.

**The construction of creating OGD user personas.** This study integrates an online user behaviors framework (Framework for Structured Research Data Interaction [30]) in the process of formulating personas. When annotating the challenges, we used the framework of human data behaviors to classify the challenges. This strategy helps contextualize user challenges and identify user characteristics associated with user behaviors, which contributes substantially to understanding users. In the meantime, the selection of the features is also considered the characteristic of OGD that requires a certain level of data literacy, which leads us to believe our persona constructing methods, including the innovative coding schema, are headed in the right direction.

**Platform Design Implications for OGD portals.** The three personas reveal that most users displayed a low proficiency in CS (*Recruits* and *Strategists*). This finding can inspire OGD platform designers to facilitate the users who are lacking the CS. For instance, designers can develop a tool that users can import the data to it and manipulate the data without the need of programming. Also, OGD portal administrators can design workshops based on user personas

to provide customized informational support, to ultimately help different groups of users. Furthermore, a regularly automatic system check is needed, especially for the downloading links. Based on the observations of the post data, we found many acquiring challenges are caused by the nonfunctional downloading links. It could vary with different portals, but regular examination should be a good suggestion for sustaining the user experience of interacting with OGD platforms.

## 6 CONCLUSION AND LIMITATION AND FUTURE WORK

This mixed methods research employed OGD online discussion forum data to investigate OGD users' characteristics and categorized the users through exploring four identified features. Ultimately, we identified three personas that help illustrate user challenges and potential needs. In addition, this work has two main limitations. While it is an initial step towards exploring OGD user characteristics, the post data we used focused on one OGD portal. However, the challenges could vary due to the different services or provided functions by portals. Another limitation concerning user proficiency coding was discussed in 3.2. Even though we agree our current method is enough to satisfy the initial analysis, we will seek alternative methods to address the possibility that users' proficiency may changed overtime with long term question answering on the forum and the case that if a user only sent one post, and to better evaluate user's proficiency in future work. Our work is a stepping stone in the understanding of OGD characteristics.

In the future, we plan to 1) collect more data to address the current limitations, and 2) invite professionals and OGD practitioners to evaluate and validate our findings.

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